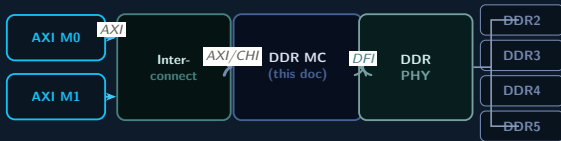


Reconfigurable Universal DDR Memory Controller

Microarchitecture Deep-Dive

DDR2 · DDR3 · DDR4 · DDR5 | AXI4 Host Interface | DFI PHY



System-level context — MC block is the scope of this document

AXI4 Host Interface

- INCR bursts only — WRAP/FIXED not used
- 64-bit data bus; outstanding multi-ID supported
- No narrow/unaligned/exclusive transfers
- Read responses ordered per AXID via ROB

DDR PHY

- Handles IO calibration, DQS alignment, ODT, ZQ cal
- Decouples MC from generation-specific electrical signalling
- Same DFI signals work for DDR3/DDR4/DDR5/LPDDR4
- Upstream boundary for this project scope

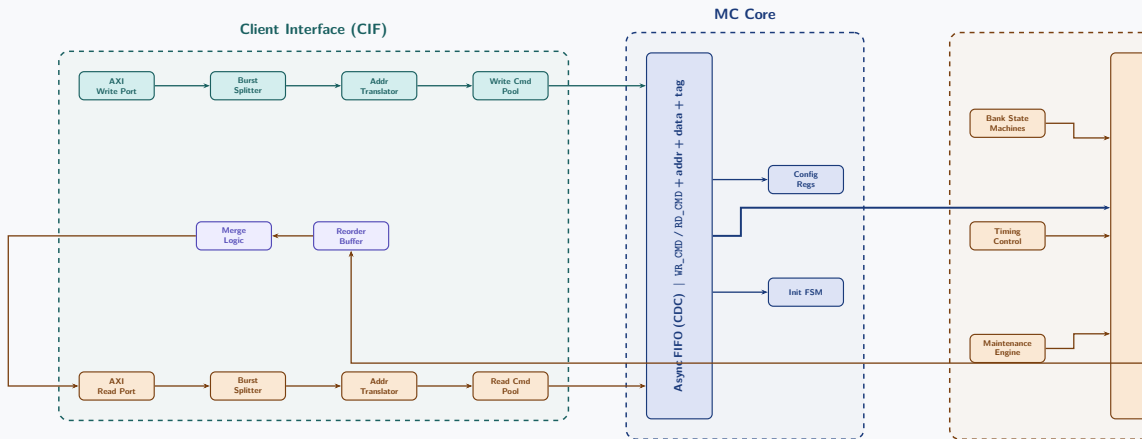
DFI (DDR PHY Interface)

- Standard MC ↔ PHY contract — MC never drives DDR pins
- DFI 5.2: unified cmd bus, per-byte DQ/DQS lanes
- MC drives dfi_address, dfi_cs_n, dfi_we_n, dfi_cas_n
- Write/read latency signalled via dfi_wrdata_en

Config Register File

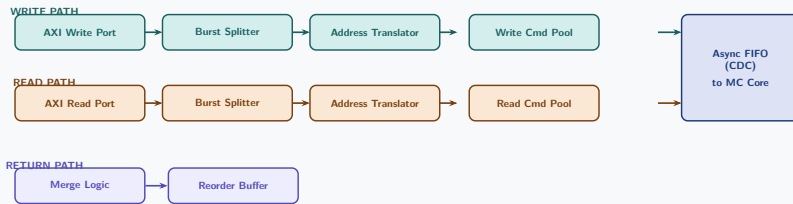
- All timing params at boot: tRCD, tRAS, tRP, tRFC, tFAW...
- Addr-mapping mode, burst length, page policy per-rank
- Same RTL correct across all supported DDR generations
- Init FSM reads command sequence from reg file

All concepts above feed directly into the MC microarchitecture discussed today.

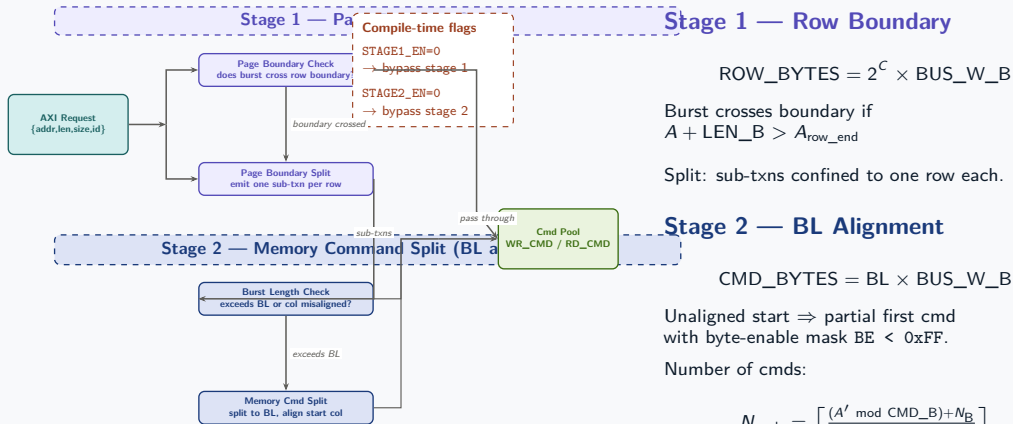


Teal = CIF Blue = MC Core Amber = DRAM Domain ▷ Today's focus: CIF (slides 4–7) + JEDEC specs (slides 9–11)

The CIF terminates the AXI4 subordinate port and converts burst transactions into a stream of normalized RD_CMD/WR_CMD packets. It has **no knowledge** of DRAM timing, bank state, or refresh.



- **Burst Splitter:** 2-stage — page boundary then BL alignment
- **Addr Translator:** byte addr → {rank, bank, row, col} via reg-file map
- **ROB:** per-AXID HEAD ptr, {AXID, seqnum} tag — no cross-ID stall
- **Merge Logic:** fragment buffer + reassembly tracker → Output Mux

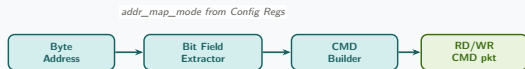


Each stage individually enable/disable at **compile time** via

`STAGE1_EN / STAGE2_EN`.

ordering

Address Translator



Output tuple: {rank, bank_grp, bank, row, col}

DDR3 No bank_grp field

DDR4 2 bank_grp bits added

DDR5 Sub-channel bits in addr

Bit-field widths are **register-programmable** — same RTL correct for all generations.

[IMAGE: addr_translator.png]

Reorder Buffer (ROB)



Property

Guarantee

Same-AXID ordering

Responses retire in strict seqnum order

Cross-AXID isolation

Per-AXID queues + HEAD ptrs independent

Tag uniqueness

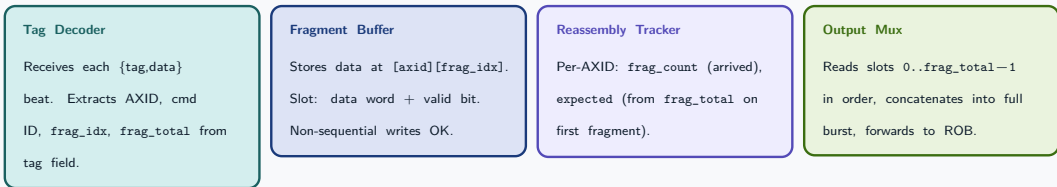
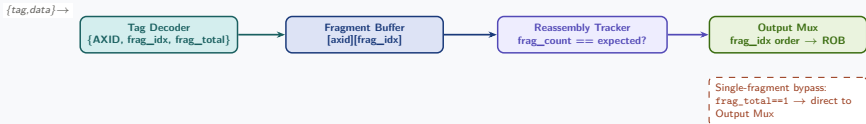
{AXID, seqnum} globally unique

CDC outside ROB

ROB on client clock only

[IMAGE: rob.png]

When the Burst Splitter splits a single AXI read into multiple sub-commands, each may return out of order from the DRAM scheduler. Merge Logic collects all fragments and concatenates them in the correct order before the ROB.



[IMAGE: merge_logic.png]

Parameter	DDR3	DDR4	DDR5
Standard	JESD79-3F	JESD79-4C	JESD79-5B
Data Rate (MT/s)	800–2133	1600–3200	3200–6400+
Prefetch	8n	8n	16n (BL16)
Burst Length (BL)	BL8 (fixed/fly)	BL8 (fixed/fly)	BL16 mandatory
Bank Groups	None	2 / 4 BGs	8 BGs (sub-ch.)
Channels	Single	Single	Sub-channel (×2)
IO Voltage	1.5 V / 1.35 V	1.2 V	1.1 V
On-Die ECC	Optional	Optional	Required (×4 org)
Refresh	tREFI = 7.8 μs	tREFI = 7.8 μs	Dual-rate refresh
Termination	RTT_NOM	RTT_PARK added	RTT_PARK + DFITMG
Key timing params	tRCD, tRP, tRAS, tWR, tRFC	tWTR_L/S added	tRFC1/2/sb, tCS

Critical Timing Parameters

tRCD	RAS-to-CAS delay; row open to col cmd
tRP	Row precharge time; min between PRE and ACT
tRAS	Min row active time ($tRCD + tRTP$)
tRC	Row cycle time = $tRAS + tRP$
tWR	Write recovery; data valid after WR before PRE
tRFC	Refresh cycle time; scales with density
tFAW	4-bank activation window; max 4 ACT in tFAW
tWTR	Write-to-read turnaround; bus direction switch
CL	CAS latency; clocks from RD cmd to first data

Mode Register Set (MRS)

- MR0: CL, BL, DLL reset, write recovery, read DQS
- MR1: DLL enable, drive strength, RTT_NOM, AL
- MR2: CWL, ASR, SRT, RTT_WR
- MR3: MPR (Multi-Purpose Register) for read training

DDR3 Initialization Sequence

1. Apply stable V_{DD} , V_{DDSPD} — min 200 μ s
2. Assert RESET# low $\geq 200 \mu$ s
3. De-assert RESET#; CKE low $\geq 500 \mu$ s
4. Apply stable CK; bring CKE high after t_{XSDLL}
5. Issue NOP $\geq t_{XPR}$ clocks
6. Load MR2 $\rightarrow$ MR3 $\rightarrow$ MR1 $\rightarrow$ MR0 (this order)
7. Issue ZQCL (ZQ calibration long) — 512 clocks
8. DRAM ready for normal operation

[IMAGE: ddr3_init_waveform.png]

Bank Group Architecture

- DDR4: 2 or 4 bank groups (BG), 4 banks each
- Intra-BG turnaround (same BG, diff bank): slower — must respect t_{RRD_L} , t_{WTR_L}
- Inter-BG turnaround (diff BG): faster — t_{RRD_S} , t_{WTR_S}
- Scheduler exploits BG interleaving for near-zero bus-turnaround penalty
- Address bit A17/BG1 selects bank group; decoded in addr translator

New Timing Parameters vs DDR3

t_{WTR_L}	Write-to-read, same bank group — longer
t_{WTR_S}	Write-to-read, different bank group — shorter
t_{RRD_L}	ACT-to-ACT same BG
t_{RRD_S}	ACT-to-ACT different BG
t_{CCD_L}	CAS-to-CAS same BG
t_{CCD_S}	CAS-to-CAS different BG

PHY Training Procedures (DDR4)

1. **Write Levelling:** DQS launched; DRAM echoes on rising edge of CK — adjusts DQS-to-CK skew per lane
2. **Read Gate Training:** PHY determines preamble window; gates DQS capture
3. **Read Centering:** Sweeps VREF on DQ bus to find centre of eye
4. **Write Centering:** PHY DQ delay tuned vs DQS
5. **ZQ Calibration:** Periodically issues ZQCS; updates R_{ZQ} impedance code

[IMAGE: ddr4_bankgroup_timing.png]

Architectural Changes vs DDR4

- **BL16 mandatory:** 16n prefetch; burst is 64 B per sub-channel
- **Sub-channels:** One DIMM = two independent 32-bit sub-channels (SA/SB); MC treats each as separate rank
- **On-Die ECC:** Required for $\times 4$ org; corrects single-bit errors before data leaves DRAM die — transparent to MC
- **8 Bank Groups:** 32 banks per sub-channel; $2\times$ parallelism vs DDR4
- **1.1V V_{DD} :** Lower swing; VDDQ separate from VDD
- **CA bus:** Command/Address bus width doubled; parity added

Dual-Rate Refresh (DDR5)

- **Normal rate:** $tREFI = 3.9\mu s$ ($2\times$ faster than DDR4)
- **Fine granularity:** Per-bank refresh (PBR) — refresh individual banks, not all; reduces refresh penalty
- **tRFC1 / tRFC2:** Two refresh modes — tRFC2 shorter, uses $2\times$ refresh rate (high-temp operation)
- **Alert mechanism:** DRAM signals MC via ALERT_n if

DDR5 Init & Training Additions

1. **BCOM training:** Sideband command bus training before main CA bus functional
2. **PDA mode:** Per-DRAM addressability — tune MRS per individual DRAM device on bus
3. **$2\times$ MRS sets:** Extended mode register space; additional latency + ODT configs
4. **RTT_PARK + DFITMG:** Runtime-switchable ODT; MC signals via DFI timing parameters
5. **CA parity:** Odd parity on CA bus; DRAM latches and validates each command frame

[IMAGE: ddr5_subchannel_arch.png]